Modern HTML5, CSS3 & Modern JavaScript ESNext Developer Ch

Category: HTML, CSS and JavaScript • Lead Mentor: Nasir • Licensed for Educational Study

1. Semantic HTML5 & Accessibility (a11y)

- Use semantic landmarks: <main>, <nav>, <header>, <footer>, <article>, and <aside>.
- Interactive elements: Always use native <button> instead of <div onclick> to maintain keyboard focus & ARIA roles.
- Accessible Forms: Explicitly pair <label for="field-id"> with input elements to support screen readers.

2. Modern CSS: Grid, Flexbox & CSS Variables

- Flexbox: Best for 1-dimensional component alignment (navbars, card headers, button groups).
- CSS Grid: Best for 2-dimensional page layouts: grid-template-columns: repeat(auto-fit, minmax(280px, 1fr)).
- CSS Custom Properties: Define tokens at :root level for scalable light/dark mode and brand theming.

3. ESNext Modern JavaScript Essentials

- Asynchronous programming: async/await with clean try/catch blocks over promise callback chains.
- Destructuring & Spread: const { id, title, ...meta } = course; const updated = { ...course, rating: 5 };
- Array Methods: Array.prototype.map(), filter(), reduce(), and some() for immutable data operations.